11.21 Controlling Mathematical Pendulum

The Pendulum Setup

Equation of motion is:

$$\ddot{\theta} = \sin\theta - \alpha\dot{\theta} + u\cos\theta$$

where:

- θ is the angle measured from the *positive* vertical direction
- u is a control
- α is the friction constant Thus u is an external horizontal force.

A physical realization

One way to implement an apparatus obeying this equation would be to place a pendulum on a moving platform. The force is the accelleration of the platform. For simplicity, the dynamics of the platform is ignored, and the interaction between the platform and the pendulum is one-directional (the platform exerts a force on the pendulum). The platform is *very heavy* as compared to the pendulum and we can move it by making it accellerate. Think of a rod mounted on top of a car, where we control the forward and reverse accelleration of the car.

The Variables

1. θ - angle with vertical (upwards) direction
2. φ - angular velocity
3. $x = (\theta, \varphi)$ - the state
4. $\lambda = (\lambda_1, \lambda_2)$ - the costate
5. u - the control
6. α - the friction constant

The Hamiltonian

We introduce the following notations:

$$\begin{aligned}
f_0 &= (1-\cos\theta) + \frac{1}{2}\varphi^2 + \frac{1}{2}u^2,\\
f_1 &= \varphi,\\
f_2 &= \sin\theta - \alpha\varphi + u\cos\theta.
\end{aligned}$$

The state equation is $\dot{x} = f(x,u)$ where $f = (f_1, f_2)$ The Hamiltonian is:

$$\begin{aligned}
\mathscr{H}(x,\lambda,u) &= f_0(x,u) + \lambda^\top \cdot f(x,u)\\
&= (1-\cos\theta) + \frac{1}{2}\varphi^2 + \frac{1}{2}u^2 + \lambda_1\varphi + \lambda_2(\sin\theta - \alpha\varphi + u\cos\theta)
\end{aligned}$$

Optimal control

⋆⋆⋆ **Step 1:** Differentiate Hamiltonian over u ⋆⋆⋆

$$\partial_u \mathscr{H} = u + \lambda_2\cos\theta$$

⋆⋆⋆ **Step 2:** Solve for u ⋆⋆⋆

$$u = u^* = -\lambda_2\cos\theta$$

⋆⋆⋆ **Step 3:** Find the *effective* Hamiltonian ⋆⋆⋆ i.e., u^* is substituted for u:

$$\begin{aligned}
H(x,\lambda) &= (1-\cos\theta) + \frac{1}{2}\varphi^2 + \frac{1}{2}\lambda_2^2\cos^2\theta + \lambda_1\varphi \\
&\quad + \lambda_2(\sin\theta - \alpha\varphi - \lambda_2\cos\theta\cdot\cos\theta) \\
&= (1-\cos\theta) + \frac{1}{2}\varphi^2 - \frac{1}{2}\lambda_2^2\cos^2\theta \\
&\quad + \lambda_1\varphi + \lambda_2(\sin\theta - \alpha\varphi).
\end{aligned}$$

Dynamical Systems view — finding stable manifold

Definition 4 (Stable Manifold)**.** If the dynamical system is $\dot{\xi} = F(\xi)$, $\xi \in \mathbb{R}^n$ then the *stable manifold*:

$$W^s(\xi_0) = \{\xi_1 \in \mathbb{R}^n \ : \ \lim_{t\to\infty} \xi(t;\xi_1) = \xi_0\}$$

where $\xi(t;\xi_1)$ is a solution such that $\xi(0,\xi_1) = \xi_1$. □

There is an *unstable manifold*:

$$W^u(\xi_0) = \{\xi_1 \in \mathbb{R}^n \ : \ \lim_{t\to-\infty} \xi(t;\xi_1) = \xi_0\}$$

The *Stable Manifold Theorem* states that the stable manifold is an actual (embedded) manifold if ξ_0 is a *hyperbolic fixed point*, i.e. $DF(\xi_0)$ has no eigenvalue with zero real part.

Mathematical basis of the algorithm

Given is the system $\dot{x} = f(x)$, $x \in \mathbb{R}^n$ and its equilibrium $x_0 = 0$. Let $\phi_t : \mathbb{R}^n \to \mathbb{R}^n$ be the *flow* of the system:

$$\phi_t(x_0) = x(t;x_0)$$

(solution of the IVP with initial condition x_0)

The flow of the system is a *1-dimensional group of diffeomorphisms*:

- $\phi_0 = id$
- $\phi_{t+s} = \phi_t \circ \phi_s$
- $\phi_{-t} = \phi_t^{-1}$

⋆⋆⋆ **Step 1:** Linearize the system at equilibrium ⋆⋆⋆ Let $A = Df(0)$

⋆⋆⋆ **Step 2:** Find eigenvalues λ_j and eigenvectors v_j ⋆⋆⋆ Suppose $\lambda_1, \lambda_2, \dots, \lambda_s$ are in the left halfplane $\mathbb{R}e\lambda < 0$.

⋆⋆⋆ **Step 3:** Parameterize the stable manifold ⋆⋆⋆ by a map $h : \mathbb{R}^s \to \mathbb{R}^n$.

⋆⋆⋆ **Step 4:** Build an auxillary system ⋆⋆⋆

$$\dot{u} = \Lambda_s u, \quad u \in \mathbb{R}^s$$

where

$$\Lambda_s = \mathrm{diag}(\lambda_1, \lambda_2, \dots, \lambda_s)$$

⋆⋆⋆ **Step 5:** conjugate the flows of the auxillary system and the main system ⋆⋆⋆

$$h(e^{t\Lambda_s}u) = \phi_t(h(u))$$

(a kind of non-linear diagonalization procedure). The idea is that near the origin h maps $\mathbb{R}^s$ to the eigenspace of all stable eigenvectors:

$$h(u) \approx \sum_{j=1}^{s} u_j v_j = V_s u$$

This leads to

$$h(u) = \phi_{-t}(h(e^{t\Lambda_s}u) \approx \phi_{-t}(V_s e^{t\Lambda_s}u)$$

as $e^{t\Lambda_s}u \to 0$, if t is large.

⋆⋆⋆ **Step 6:** Take t to infinity ⋆⋆⋆ The final computational formula is:

$$h(\mathbf{u}) = \lim_{t\to\infty} \phi_{-t}\left(\sum_{j=0}^{s} e^{\lambda_j t} v_j\right)$$

MATLAB implementation for the pendulum control problem

```
clear all;
%% Set up the Hamiltonian system symbolically
syms x y u a H l0 l1 l2 f0 f1 f2

% Cost function
f0 = 1/2*y*y + 1/2*u*u + 1-cos(x);
f1 = y;
f2 = sin(x) + u*cos(x) - a*y;
H = f0 + l1*f1 + l2*f2;

ustar = solve(diff(H, u),u);
H = subs(H, {a, u}, {0.1, ustar});

vars = { x, y, l1, l2 };

% Find the Hessian of Hamiltonian at 0
HessianSymbolic = hessian(H, [vars{:}]);
HessianAtZero = subs(HessianSymbolic,vars,{0,0,0,0});
HessianNumeric = double(HessianAtZero);

% Note V=W; left and right eigenvectors are the same
J = [zeros(2,2), eye(2); - eye(2),zeros(2,2)];

% Find the "Hamiltonian matrix"
C = J*HessianNumeric;

%% Solve ARE
% C*V = V*D; W'*C = D*C'.
[V,D,W] = eig(C);
D = diag(D);
ind = D < 0;
D = D(ind);
Vs = V(:,ind);
Vs1 = Vs(1:2,:);
Vs2 = Vs(3:4,:);
P = Vs2*inv(Vs1);

% Full Hamiltonian vector field
F(x,y,l1,l2) = [diff(H,l1), diff(H,l2), -diff(H,x), -diff(H,y)];

% Numerical hamiltonian in a form suitable for ODE solver; need dummy time
G = @(~, z) double(F(z(1),z(2),z(3),z(4)))';

Tspan = linspace(0,7,100);

x0 = [0.01,0]';
y0 = P*x0;
z0 = [x0;y0];

%% Draw trajectories, using ode23 and ode45 for comparison
[t, z23] = ode23(G, Tspan, z0);
[t, z45] = ode45(G, Tspan, z0);
clf;
hold on;
plot(z23(:,1),z23(:,2));
plot(z45(:,1),z45(:,2));
```

```
%xlim([-pi,pi])
%ylim([-2,2])
hold off;


% Shooting method - the spider web
psi = linspace(0,2*pi,100);
rad = 5;
% Write x as a linear combination of stable eigenvectors
tmax = 20;

% Solve linear equation forward, starting initial conditions on a circle
z = Vs * expm(tmax*diag(D)) * rad * [cos(psi); sin(psi)];

% Visualize
%x = z(1:2,:);
%plot(x(1,:),x(2,:),'-');


% Reverse time vector field
Grev = @(~, z) -double(F(z(1),z(2),z(3),z(4)))';

% Negotiate a time step
target_time_step = .2;
num_timesteps = round( tmax / target_time_step);
timestep = tmax / num_timesteps;

num_anglesteps = size(z,2);

Tspan = linspace(0, tmax, num_timesteps);
Z = zeros(num_anglesteps, num_timesteps, 4);

% Computational loop: Replace 'parfor' with 'for' to
% run single-threaded.
tic;
parfor j=1:num_anglesteps
    z_start = z(:,j);
    [t, z_traj] = ode45(Grev, Tspan, z_start);
    Z(j,:,:) = z_traj;
end
toc;

% Visualization loop
clf;
hold on;
for j=1:num_anglesteps
    z_traj = squeeze(Z(j,:,:));
    %plot(z_traj(:,1),z_traj(:,2));
    plot3(z_traj(:,1),z_traj(:,2),z_traj(:,3),'-o');
    drawnow;
end
for l=1:num_timesteps
    z_traj = squeeze(Z(:,l,:));
    %plot(z_traj(:,1),z_traj(:,2));
    plot3(z_traj(:,1),z_traj(:,2),z_traj(:,3),'-o');
    drawnow;
end
xlabel("\theta");
ylabel("\phi");
zlabel("\lambda_1");
% Found interactively
view(82,-18);
title("Stable manifold parameterized by angle and time");
hold off;
saveas(gcf,'StableManifoldParameterization.png')
```

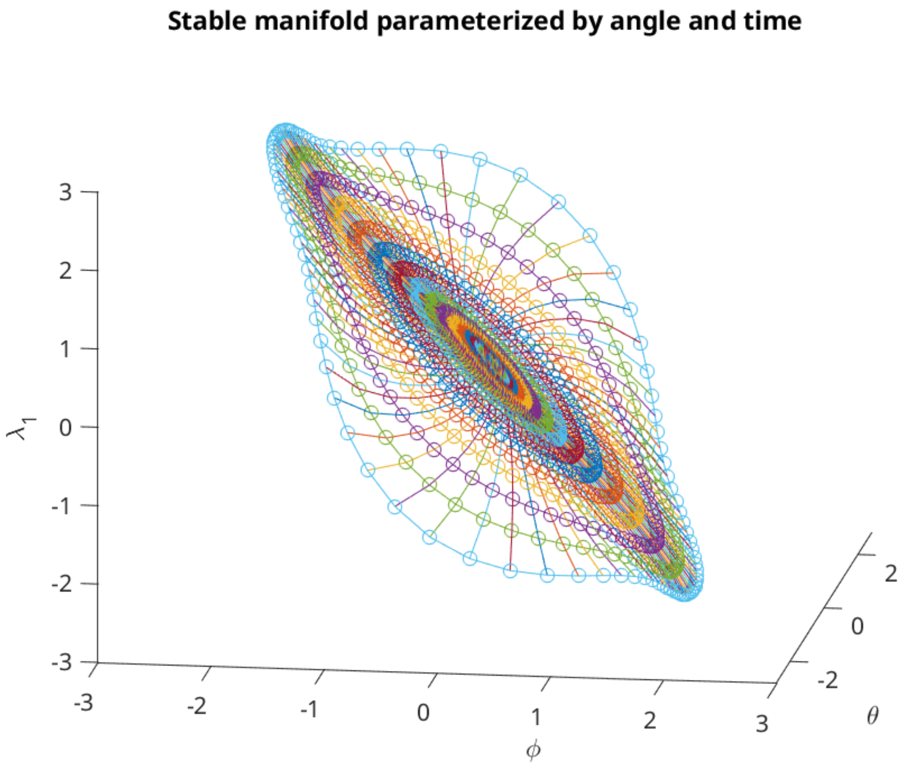

Figure 11.3: The stable manifold projected on three dimensions

A view of the stable manifold is provided in Figure 11.3.

11.22 Exercises

11.22.1 Linear-Quadratic Regulator (LQR)

Exercise 11.1 (Optimal control and stability analysis)**.** Consider the scalar system $\dot{x}(t) = 2x(t) + 3u(t)$, with cost

$$J = \int_0^\infty (4x^2(t) + 2u^2(t))\, dt.$$

Derive the optimal control $u^*(t)$ and analyze the stability. □

Exercise 11.2 (Stability conditions)**.** Given the system $\dot{x}(t) = -x(t) + 2u(t)$ and cost

$$J = \int_0^\infty (x^2(t) + u^2(t))\, dt,$$

determine the equations of motion and conditions for stability. □

Exercise 11.3 (Effective Hamiltonian)**.** Consider the Hamiltonian system

$$H(x, u, \lambda, t) = x^2 + u^2 + \lambda(4x + u).$$

Find the optimal control u^* and effective Hamiltonian. □